Algebraic endpoint densities are the transfer-matrix charges in regular Yang–Baxter chains

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Abstract

Schouten and de Leeuw recently introduced algebraic charge densities obtained by differentiating directed products of a regular Yang–Baxter R-matrix and conjectured that two endpoint differences represent the usual logarithmic-transfer charges. We prove this conjecture for every finite-dimensional local space, every regular braiding-normalized (not necessarily difference-form) R-matrix, every charge order $k \geq 2$, and every homogeneous untwisted periodic chain of length $L \geq k$. Regularity unfolds a multivariable normalized transfer matrix at any inactive edge. Coefficients of its logarithmic derivative are then rooted connected clusters, with a degree-r coefficient supported on at most $r+1$ edges. Subtracting the shortened open product anchors each cluster once, so the cyclic sums of both directed endpoint densities equal the prescribed transfer charge, including at the minimal size $L = k$. Finally, an explicit tensor-word argument identifies the kernel of cyclic summation: every vanishing k-site density is exactly a local gradient $\mathbf{1} \otimes m - m \otimes \mathbf{1}$. This upgrades equality of global charges to the precise local equivalence in the conjecture, without a thermodynamic limit, boost boundary terms, or removal of lower-charge admixtures.

1 Introduction

A regular solution of the Yang–Baxter equation supplies a commuting transfer matrix and, by logarithmic differentiation, a hierarchy of local conserved charges. The hierarchy is central to quantum-integrable spin chains, while practical constructions often use boost or ladder recurrences instead of differentiating a long monodromy directly [1, 2, 3, 4, 5]. For a general non-difference R-matrix, however, the choice of local density and its relation to an algebraic representative are delicate: equal periodic sums determine a density only modulo a local gradient, and scalar or lower-range terms cannot simply be discarded.

Schouten and de Leeuw introduced two directed families of algebraic charge densities in their quantum-group treatment of long-range deformations [6]. Their Conjecture 2.7 asserts that a difference at the two ends of either directed block is locally equivalent to a density of the usual logarithmic-transfer charge. They established the first orders and the commutation of the candidate sums with the transfer matrix, but left their identification with the specified charge hierarchy open. The current arXiv version at the time of writing is v2, dated 5 June 2026; a bounded exact-title, identifier, author and conjecture-phrase search on 14 October 2026 found no later primary proof or disproof. This search is not an exhaustive priority claim.

Here we prove the conjecture in its undeformed scope. The proof has three ingredients. First, regular permutation factors turn the two algebraic stars into the left and right Maurer–Cartan forms of one open nearest-neighbour product. Second, cutting a multivariable

transfer matrix at a regular factor yields an exact product over active connected components. Its rooted logarithmic coefficients are therefore connected and have a sharp support bound. Endpoint subtraction anchors each such coefficient exactly once. Third, a constructive tensor-word lemma shows that the kernel of cyclic summation is precisely the space of local gradients. Yang–Baxter integrability is used only to equate the left and right logarithmic derivatives on the physical homogeneous line.

The result concerns dimensionless raw derivatives at undeformed order. It does not extend by itself to twisted, open or inhomogeneous chains, nor to positive orders of a long-range deformation. In a Hermitian realization, conventional factors of i and an energy scale must still be restored model by model.

2 Setup and statement

Let V be a finite-dimensional complex vector space and let

$$R(u,v) \in \mathrm{End}(V \otimes V)[[u,v]] \tag{1}$$

be a formal matrix power series, or the Taylor series of an analytic matrix near $(0,0)$. We assume the Yang–Baxter equation

$$R_{12}(u_1,u_2)R_{13}(u_1,u_3)R_{23}(u_2,u_3) = R_{23}(u_2,u_3)R_{13}(u_1,u_3)R_{12}(u_1,u_2), \tag{2}$$

regularity $R(u,u) = P$, and the fixed braiding normalization

$$R_{12}(u,v)R_{21}(v,u) = \mathbf{1}. \tag{3}$$

Equation (3) is not Hilbert-space unitarity. No difference-form, Hermiticity or diagonalizability hypothesis will be used.

On a homogeneous periodic chain of length L, define

$$t_L(u) = \mathrm{tr}_a[R_{aL}(u,0)\cdots R_{a1}(u,0)]\,, \qquad U_L = t_L(0), \tag{4}$$

and the raw logarithmic-transfer charges

$$Q_k^{(L)} = \partial_u^{k-2}\big(t_L(u)^{-1}t_L'(u)\big)\Big|_{u=0}\,, \qquad k \geq 2. \tag{5}$$

Regularity makes U_L the invertible one-site translation. There is no factorial division in (5).

For ordered disjoint blocks $A = (1,\ldots,N)$ and $B = (N+1,\ldots,M)$, use the directed product convention of Ref. [6],

$$R_{AB} = \prod_{i=1}^{N}\prod_{j=M}^{N+1} R_{ij}(u_i,u_j), \qquad D_A = \sum_{i\in A}\partial_{u_i}, \quad D_B = \sum_{j\in B}\partial_{u_j}, \tag{6}$$

where the displayed order is the operator order. The algebraic densities are

$$\mathcal{Q}_{AB}^{(k)} = D_A^{k-2}\left(R_{AB}^{-1}D_A R_{AB}\right), \tag{7}$$

$$\widetilde{\mathcal{Q}}_{AB}^{(k)} = -D_B^{k-2}\left(R_{AB}^{-1}D_B R_{AB}\right). \tag{8}$$

All spectral parameters are set to zero after differentiation. Empty-block terms are zero. Two k-site densities are locally equivalent, written $x \sim y$, when

$$y - x = \mathbf{1}\otimes m - m\otimes\mathbf{1} \tag{9}$$

for some $m \in \mathrm{End}(V^{\otimes(k-1)})$.

Theorem 2.1 (Endpoint-density identity)**.** *Under* (2)–(3)*, for every* $k \geq 2$ *the two endpoint differences*

$$a_k = \left(\mathcal{Q}^{(k)}_{1(2,\dots,k)} - \mathcal{Q}^{(k)}_{2(3,\dots,k)}\right)\Big|_{\boldsymbol{u}=0}, \tag{10}$$

$$b_k = \left(\widetilde{\mathcal{Q}}^{(k)}_{(1,\dots,k-1)k} - \widetilde{\mathcal{Q}}^{(k)}_{(1,\dots,k-2)k-1}\right)\Big|_{\boldsymbol{u}=0} \tag{11}$$

are locally equivalent. On every homogeneous untwisted periodic chain $L \geq k$*, each cyclic sum is exactly* $Q^{(L)}_k$*. Consequently, both are equivalent in the sense of* (9) *to any* k*-site density of the charge* (5)*.*

3 The directed blocks as open logarithmic derivatives

Put

$$C(u) = PR(u,0), \qquad C(0) = \mathbf{1}, \tag{12}$$

and, on sites $1,\dots,n$, define

$$G_n(u) = C_{n-1,n}(u)\cdots C_{12}(u), \qquad G_1 = \mathbf{1}, \tag{13}$$

together with the left and right logarithmic derivatives

$$F_n = G_n^{-1}G_n', \qquad B_n = G_n'G_n^{-1}. \tag{14}$$

Superscripts on F_n or B_n will indicate a shifted interval of sites.

Proposition 3.1 (Regular-factor reduction)**.** *For* $r = k-2$*, the source expressions* (10)–(11) *are*

$$a_k = \partial_u^r \left(F_k - F_{k-1}^{[2,k]}\right)\Big|_{u=0}, \tag{15}$$

$$b_k = \partial_u^r \left(B_k - B_{k-1}^{[1,k-1]}\right)\Big|_{u=0}. \tag{16}$$

Proof. Let

$$S_n(u) = R_{1n}(u,0)\cdots R_{12}(u,0), \qquad W_n = P_{1n}\cdots P_{12}. \tag{17}$$

Moving each regular flip to the left gives

$$W_n^{-1}S_n(u) = G_n(u). \tag{18}$$

This follows inductively: after extracting P_{1n}, conjugation by W_{n-1} carries C_{1n} to $C_{n-1,n}$ without reversing its tensor slots. Since W_n is independent of u, $S_n^{-1}S_n' = F_n$. Taking the full and shortened stars yields (15).

For the opposite orientation, (3) gives, in fixed tensor slots,

$$R_{ij}(0,u) = R_{ji}(u,0)^{-1} = P_{ij}C_{ij}(u)^{-1}. \tag{19}$$

If $T_n(u) = R_{1n}(0,u)\cdots R_{n-1,n}(0,u)$, moving the flips through the factors to their right gives

$$T_n(0)^{-1}T_n(u) = C_{12}(u)^{-1}\cdots C_{n-1,n}(u)^{-1} = G_n(u)^{-1}. \tag{20}$$

Therefore

$$-T_n^{-1}T_n' = -G_n(G_n^{-1})' = G_n'G_n^{-1} = B_n, \tag{21}$$

which retains the single minus sign in (8). Full minus shortened blocks now give (16). No difference property was used. □

4 Cutting the multivariable transfer

Introduce independent commuting bookkeeping variables $z_1, \dots, z_L$ and the normalized multivariable transfer

$$X_L(\boldsymbol{z}) = U_L^{-1} \operatorname{tr}_a[R_{aL}(z_L, 0) \cdots R_{a1}(z_1, 0)]\,. \tag{22}$$

This is only a generating series. We do not assert that it is a new commuting inhomogeneous family. On the diagonal $z_1 = \cdots = z_L = u$, $X_L = U_L^{-1} t_L(u)$. Associate z_j with the directed edge $(j-1, j)$, with indices understood cyclically.

Lemma 4.1 (Exact cut and component factorization)**.** *If S is a proper subset of the L directed edges and all variables outside S are set to zero, then*

$$X_L|_S = \prod_{I \in \pi_0(S)} G_I, \tag{23}$$

where I runs over the connected interval components of S and G_I is the corresponding open product (13)*. Products belonging to different components commute.*

Proof. Choose an inactive edge, say $z_1 = 0$. Regularity inserts $R_{a1}(0, 0) = P_{a1}$. The elementary partial-trace identity

$$\operatorname{tr}_a(A_a P_{a1}) = A_1 \tag{24}$$

replaces the auxiliary slot by site 1. Equation (18), with an independent variable in each factor, then gives

$$X_L(0, z_2, \dots, z_L) = C_{L-1,L}(z_L) \cdots C_{12}(z_2). \tag{25}$$

A cut at any other inactive edge follows by cyclically rotating the monodromy before the partial trace. This use of partial-trace cyclicity is valid because physical matrix entries of the rotated factor commute with those on all other sites. Additional inactive edges split (25) into open products on disjoint site intervals. Those factors have disjoint support and hence commute, proving (23) and independence of the chosen cut. □

Define the rooted logarithmic derivatives

$$Y_j = X_L^{-1} \partial_{z_j} X_L, \qquad Z_j = (\partial_{z_j} X_L) X_L^{-1}. \tag{26}$$

For a coefficient of the monomial $\boldsymbol{z}^{\boldsymbol{\alpha}}$ in either series, its rooted edge support is

$$S(\boldsymbol{\alpha}, j) = \{i : \alpha_i > 0\} \cup \{j\}. \tag{27}$$

The root j is included even if $\alpha_j = 0$, since differentiation may have removed its only power.

Proposition 4.2 (Rooted connectedness and range)**.** *A degree-r coefficient in Y_j or Z_j has rooted support on at most $r+1$ edges. If $r \le L-2$, every nonzero coefficient has connected rooted support.*

Proof. The cardinality bound follows immediately from (27). It makes the support proper when $r \le L-2$, so Lemma 4.1 computes the coefficient exactly after all other variables are set to zero. If the support is disconnected, let I be the component containing the root. Differentiating (23) and multiplying by the product inverse gives

$$(X_L^{-1} \partial_{z_j} X_L)|_S = G_I^{-1} \partial_{z_j} G_I, \qquad ((\partial_{z_j} X_L) X_L^{-1})|_S = (\partial_{z_j} G_I) G_I^{-1}. \tag{28}$$

Neither expression contains a variable belonging to another component. Since every non-root edge in S has a positive exponent, the proposed disconnected coefficient must vanish. □

5 Endpoint anchoring and the global charge

Let τ translate a local operator by one site and write the cyclic sum of a k-site density d on length L as

$$\mathsf{S}_L[d] = \sum_{n=0}^{L-1} \tau^n(d). \tag{29}$$

Proposition 5.1 (Each connected coefficient is counted once)**.** *For $r = k-2$ and $L \geq k$,*

$$\mathsf{S}_L[a_k] = \partial_u^r \big(X_L(u)^{-1} X_L'(u)\big)\Big|_{u=0}, \tag{30}$$

$$\mathsf{S}_L[b_k] = \partial_u^r \big(X_L'(u) X_L(u)^{-1}\big)\Big|_{u=0}. \tag{31}$$

Proof. By Proposition 4.2, a surviving degree-r rooted coefficient occupies a connected interval of at most $r+1 = k-1$ edges. It is therefore nonwrapping even at the minimal length $L = k$. In the left form, subtracting $F_{k-1}^{[2,k]}$ from F_k cancels exactly those rooted coefficients whose support omits the first edge. This remains true when the removed edge is the root and its exponent after differentiation is zero: such a term has no counterpart in the shortened product. Thus every remaining interval is anchored at its unique leftmost edge. Cyclic translation counts every connected rooted coefficient exactly once.

For the right form, subtracting $B_{k-1}^{[1,k-1]}$ instead anchors the unique rightmost edge, and the same argument applies. Raw differentiation multiplies every degree-r coefficient by the same $r!$ on both sides. The case $k = 2$ is included: the rooted support consists only of its root and $G_1' = 0$. □

The Yang–Baxter equation now enters. Its standard train argument gives $[t_L(u), t_L(v)] = 0$. Hence t_L, t_L' and $U_L = t_L(0)$ commute, and on the homogeneous line

$$X_L^{-1} X_L' = t_L^{-1} t_L' = X_L' X_L^{-1}. \tag{32}$$

Combining Proposition 5.1 with (32) proves

$$\mathsf{S}_L[a_k] = Q_k^{(L)} = \mathsf{S}_L[b_k] \qquad (L \geq k). \tag{33}$$

No infinite-chain limit or finite periodic boost is involved, and scalar and lower-range coefficients remain in the calculation.

6 The exact kernel of cyclic summation

Global equality is not yet the local equivalence (9). The required upgrade follows from a finite-dimensional algebraic lemma.

Lemma 6.1 (Cyclic kernel)**.** *Let $D \in \mathrm{End}(V^{\otimes k})$. If $\mathsf{S}_L[D] = 0$ for one $L > 2k$, then there exists $M \in \mathrm{End}(V^{\otimes(k-1)})$ such that*

$$D = \mathbf{1} \otimes M - M \otimes \mathbf{1}. \tag{34}$$

Conversely, every expression of the form (34) *has zero cyclic sum for every $L \geq k$.*

Proof. Choose a basis of $\mathrm{End}(V)$ consisting of $\mathbf{1}$ and trace-zero matrices E_α, and expand D in the induced tensor-word basis. For any word other than the all-identity word, delete

its leading and trailing identities. The remaining *core* w has length $m \le k$ and nonidentity first and last letters. Let c_a be the coefficient of the embedding

$$\mathbf{1}^{\otimes a} \otimes w \otimes \mathbf{1}^{\otimes(k-m-a)}, \qquad 0 \le a \le k-m. \tag{35}$$

On a ring with $L > 2k$, the exterior identity run uniquely identifies the core and its cyclic placement, so distinct cores cannot cancel one another. Vanishing of the cyclic sum is therefore equivalent to a zero scalar coefficient and

$$\sum_{a=0}^{k-m} c_a = 0 \tag{36}$$

for every core.

For $m < k$, put the same core into a $(k-1)$-site word and assign coefficients

$$d_a = -\sum_{b=0}^{a} c_b, \qquad 0 \le a \le k-m-1. \tag{37}$$

Then $\mathbf{1} \otimes M - M \otimes \mathbf{1}$ has coefficients $-d_0 = c_0$, $d_{a-1} - d_a = c_a$ in the interior, and $d_{k-m-1} = c_{k-m}$ at the final embedding, where the last equality is (36). Summing these contributions over all cores constructs M. For $m = k$, (36) forces the sole coefficient to vanish. If $\dim V = 1$, only the scalar word exists and is likewise forced to zero. The converse is immediate by telescoping around the ring. □

Apply Lemma 6.1 to $D = a_k - b_k$, using (33) at any $L > 2k$. This proves $a_k \sim b_k$. If q_k is any k-site representative whose cyclic sum is $Q_k^{(L)}$ for all $L \ge k$, the same lemma applied to $a_k - q_k$ and $b_k - q_k$ proves both equivalences claimed in Theorem 2.1. Alternatively, a_k itself supplies a canonical representative, so no independently normalized density need be assumed. This completes the proof of Theorem 2.1.

7 Exact finite checks and convention control

The theorem is analytic; the following computations test conventions rather than supply its universal quantifiers. Two independent implementations transcribed the transfer matrix and both directed block products using exact integer or rational arithmetic. Both implementations were developed with OpenAI Codex assistance; their independence concerns the input reconstruction and matrix representation, not independent human authorship. Generated code was checked by complete exact residuals and a separate fresh AI reconstruction. The published spin-$\frac{1}{2}$ XXX input

$$R_{12}(u,v) = \frac{(u-v)\mathbf{1} - P_{12}}{u-v-1} \tag{38}$$

reproduces the known Q_2, Q_3, Q_4 densities after the explicit normalization map to Ref. [4]. A nonlinear reparametrization $f(u) = u+u^2$ retains the chain-rule admixtures $Q_4[f] = Q_4+6Q_3$ and $Q_5[f] = Q_5 + 12Q_4 + 12Q_3$; both endpoint formulas solve the complete local-gradient system at $k = 4, 5$. An admissible spectral-gauge transform provides a separate non-difference control through $k = 6$, including $L = k = 6$.

The recorded exact checks are as follows; "zero" means entrywise exact zero, with no floating-point tolerance.

- For XXX at $k \le 4$ and $L = 4, 5, 6$, independent transfer and block constructions have zero residual.

- For $f(u) = u + u^2$ at $k = 4$ with $L = 4, 5, 6$ and at $k = 5$ with $L = 5, 6$, the independent transfers agree exactly; both complete local-gradient systems also have zero residual, and hence the endpoint sums equal the same charge.
- For spectral-gauge XXX at $k = 4, 5, 6$ and $L = 6, 7$, the sums, gradients and independent transfer comparison have zero residual.
- For generic non-Yang–Baxter controls through $k = 5$, the cut and connected-support identities have zero residual.

The generic controls in the last row are deliberately outside the physical target and are not counterexamples or confirmations of the Yang–Baxter theorem. They isolate the purely algebraic cut and support steps. Reproduction commands, scripts, exact outputs and the Python version are specified in the accompanying reproducibility record.

8 Physical meaning and limitations

The theorem identifies the algebraic densities of Ref. [6] with the particular hierarchy selected by the logarithmic derivative (5); they are not merely another mutually commuting family. In particular, no untracked scalar or lower-charge combination remains after passage to the exact local-gradient quotient. This supplies the missing undeformed identification used conditionally in that work and gives finite-support representatives at every order.

For a Hermitian XXX realization one may write $H = JQ_2$ with real J carrying energy units. Odd raw derivatives can require the customary factor of i to become Hermitian. The general algebraic R in Theorem 2.1 need not be a unitary quantum gate, and the theorem does not predict a new excitation, phase or experimental signal. Its scope is the homogeneous, untwisted, periodic, undeformed chain. Open boundaries, twists, inhomogeneous evaluation parameters and positive long-range deformation orders can introduce additional endpoint or wrapping terms and are not covered.

9 Conclusion

Regularity turns an inactive transfer factor into an exact cut. This makes the logarithmic-transfer coefficients rooted connected clusters and supplies a support bound strong enough to retain the minimal chain length $L = k$. The two directed endpoint subtractions anchor those clusters from opposite ends, while Yang–Baxter commutativity identifies the resulting left and right logarithmic derivatives. A constructive cyclic kernel lemma then yields the precise local gradient. Together these steps prove both orientations of Conjecture 2.7 of Ref. [6] for general regular braiding-normalized R-matrices and all charge orders.

Data and code availability

The exact-arithmetic scripts, machine-readable outputs, primary-source locators and regeneration commands accompanying this local draft are described in `REPRODUCIBILITY.md`. They test finite cases and conventions; the all-order claim rests on the proof above. No third-party article PDF is part of the proposed submission bundle.

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OpenAI Codex was used to support literature retrieval and organization, proof development, exact-arithmetic code, an independent AI audit, and manuscript drafting. AI-assisted outputs were checked by exact replays and by a separate fresh AI scientific review, but those checks are not external peer review. This is a local draft: the named author must critically review and validate all content, approve the disclosure, and take responsibility for any version submitted to a journal.

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